

MAT1100 Homework 1

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Question 4

- (a) *Proof.* To show H is not normal in G , consider $r^{-1} \in G$ and $rs \in H$ then

$$r^{-1}(rs)r = sr = r^{-1}s$$

which is not in H , hence H is not invariant under conjugation meaning it is not normal. To show that

$$(aH)(bH) = (ab)H$$

is not a well defined operation, it suffices to show that given a', b' , other representatives of aH and bH

$$(a'b')H \neq (ab)H$$

to do this take $a = b = s$ and $a' = b' = s \cdot rs$ which is also a representative of sH as $rs \in H$. Now note that

$$ab = s^2 = e$$

and

$$a'b' = s \cdot rs \cdot s \cdot rs = sr^2s = r^{-2}s^2 = r^2$$

and clearly $r^2 \notin H$ therefore

$$r^2H \neq H$$

which implies the above operation is not well defined. □

- (b) *Proof.* To show that N is normal recall that by question 3, $N = Z(D_8)$ and therefore for all $x \in D_8$ and $g \in N$

$$xg = gx \implies xgx^{-1} = g \in N$$

thus N is normal. The cosets of N in G are:

$$N = \{e, r^2\}$$

$$rN = \{r, r^3\}$$

$$sN = \{s, sr^2\}$$

$$srN = \{sr, sr^3\}$$

□

- (c) *Proof.* If D_8/N were cyclic, it would be generated by a single element of order $|D_8|/|N| = 4$. Thus to show D_8/N is not cyclic, it suffices to show that it contains no element of order greater than 2. To show that a given element $aN \in D_8/N$ has order at most 2, it suffices to pick a representative g of aN and show that $g^2 \in N$. By part (b) the cosets of N in D_8 are

$$\{N, rN, sN, srN\}$$

and fix representatives

$$\{e, r, s, sr\}$$

then

$$\begin{aligned} e^2 &= e \in N \\ (r)^2 &= r^2 \in N \\ s^2 &= e \in N \\ (sr)^2 &= (sr) \cdot (r^{-1}s) = s^2 = e \in N \end{aligned}$$

hence all elements of D_8/N have order at most 2 and thus the group cannot be cyclic as it has order 4.

Now to prove that D_8/N is abelian since every element of D_8/N squares to the identity, it suffices to show that for any group G if $x^2 = e$ for all $x \in G$ then G is abelian. To do this, note that $x^2 = e$ is equivalent to $x = x^{-1}$ and therefore if $a, b \in G$

$$ab = a^{-1}b^{-1} = (ba)^{-1} = ba$$

where $(ba)^{-1} = ba \in G$ from the fact that every element in G squares to the identity. Hence G is abelian and therefore D_8/N is abelian. $\square$